Zecpath Technical Interview AI Blueprint

Objective

Design a scalable AI-powered technical interview system capable of conducting adaptive, role-based technical interviews for multiple job domains.

System Overview

The Technical Interview AI evaluates candidates by selecting questions based on:

- Job Role
- Experience Level
- Skill Domain
- Previous Answers
- Difficulty Progression

The system dynamically adjusts question complexity throughout the interview.

Core Components

1. Role & Skill Mapper

Maps every job role to its required technical skills.

Examples:

- MERN Developer → JavaScript, React, Node.js, MongoDB
- Java Developer → Java, Spring Boot, Microservices
- DevOps Engineer → Docker, Kubernetes, CI/CD
- Data Scientist → Python, Machine Learning, Statistics

---

2. Question Engine

Generates technical questions based on:

- Candidate Role
- Experience
- Skill
- Current Difficulty

---

3. Difficulty Engine

Adjusts interview difficulty according to candidate performance.

- Good Answer → Increase difficulty
- Average Answer → Maintain level
- Poor Answer → Reduce difficulty

---

4. Answer Evaluation Engine

Evaluates:

- Technical correctness
- Logical reasoning
- Code quality
- Explanation depth

---

5. Follow-Up Engine

Creates contextual follow-up questions based on previous answers.

---

6. Technical Scoring Engine

Scores candidates using:

- Accuracy
- Technical Depth
- Optimization Thinking
- Communication

---

7. Report Generator

Generates:

- Technical Score
- Strengths
- Weaknesses
- Hiring Recommendation

Overall Workflow

Start Interview

↓

Role Detection

↓

Experience Classification

↓

Skill Mapping

↓

Question Generation

↓

Answer Evaluation

↓

Difficulty Adjustment

↓

Technical Scoring

↓

Report Generation

↓

End Interview

Advantages

- Adaptive Interviewing
- Role-based Questions
- Scalable Architecture
- Explainable Evaluation
- Modular Components

Future Improvements

- Live Coding Evaluation
- LLM-based Answer Analysis
- IDE Integration
- Code Execution Sandbox